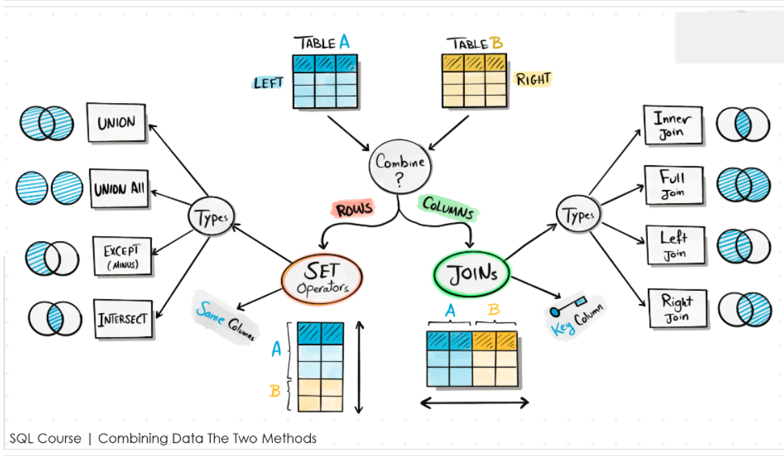
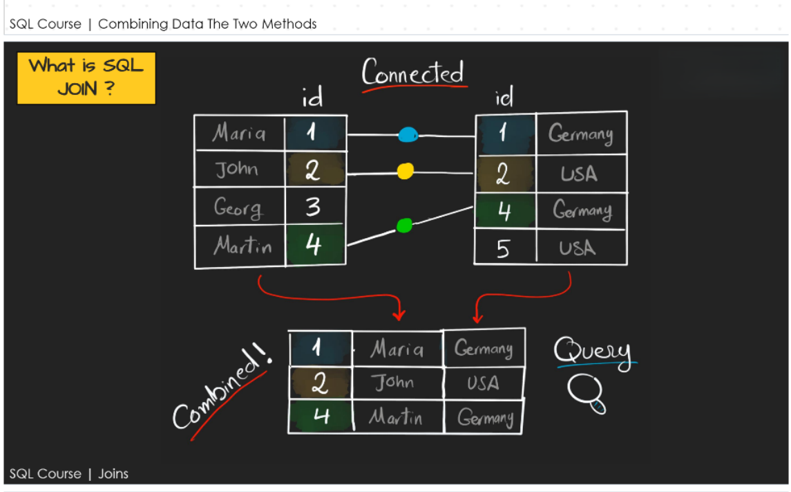
SQL JOINS

Combining Data



SQL Course | Combining Data The Two Methods



	id
Maria	1
John	2
Georg	3
Martin	4

id	
1	Germany
2	USA
4	Germany
5	USA

1	Maria	Germany
2	John	USA
4	Martin	Germany

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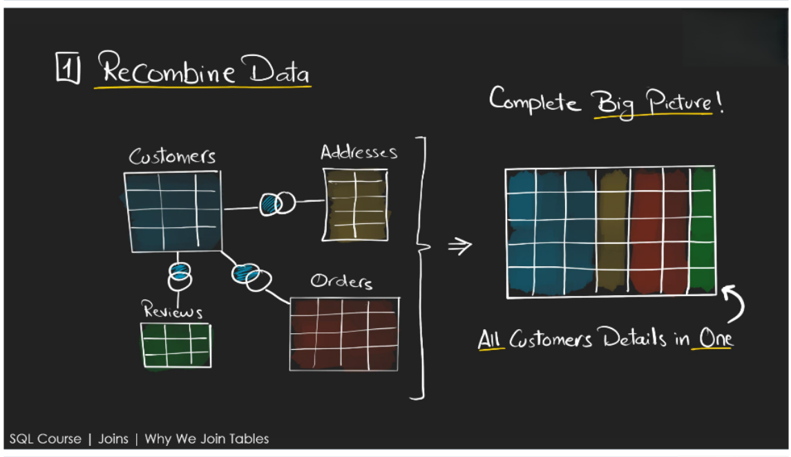
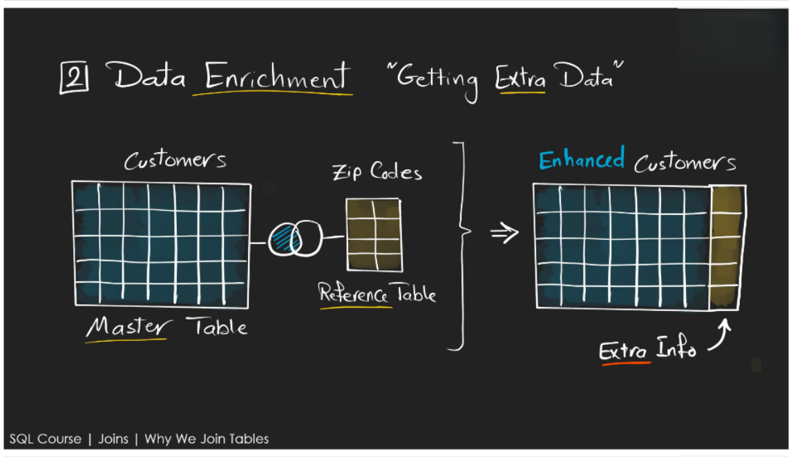
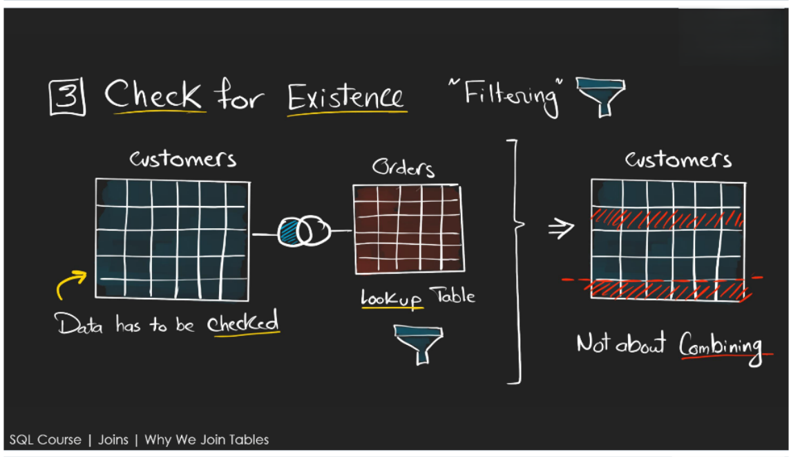


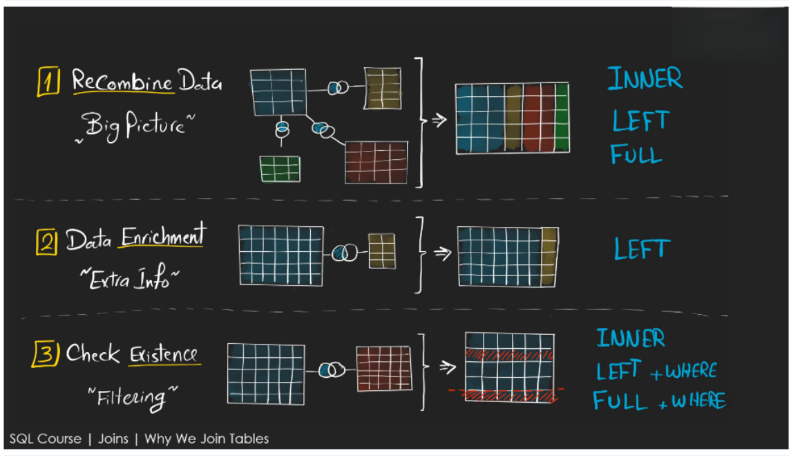
SQL Course | Joins | Why We Join Tables



SQL Course | Joins | Why We Join Tables

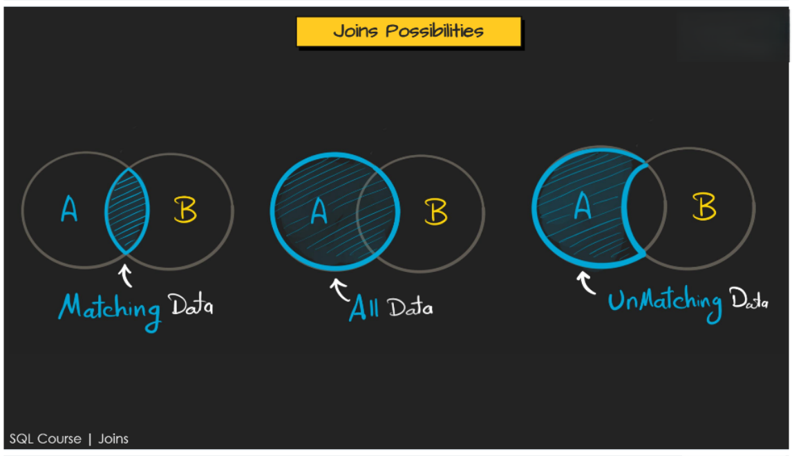


SQL Course | Joins | Why We Join Tables



1 Recombine Data "Big Picture"	INNER LEFT FULL
2 Data Enrichment "Extra Info"	LEFT
3 Check Existence "Filtering"	INNER LEFT + WHERE FULL + WHERE

SQL Course | Joins | Why We Join Tables



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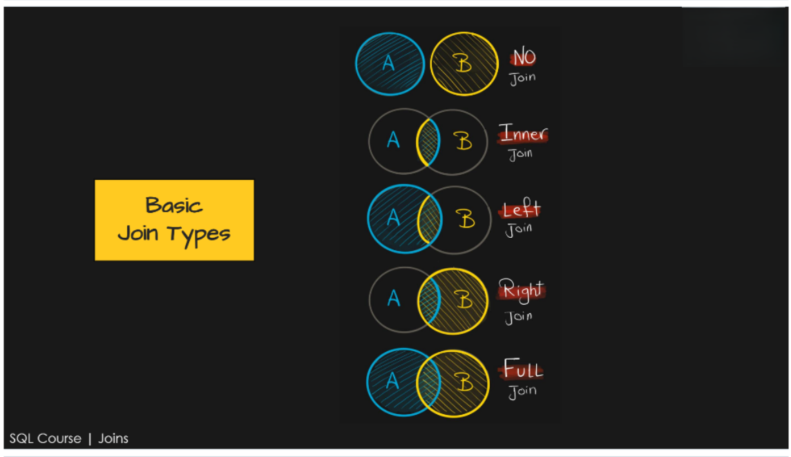

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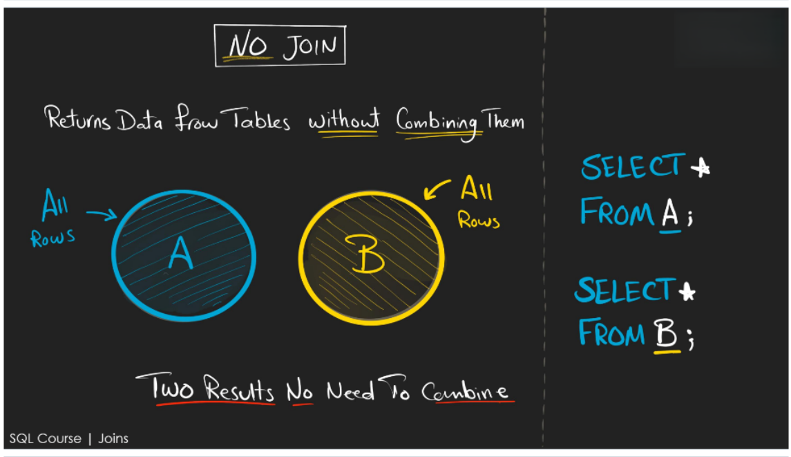

Returns Data from Tables without Combining Them

Two Results No Need To Combine

```
SELECT *
FROM A;

SELECT *
FROM B;
```

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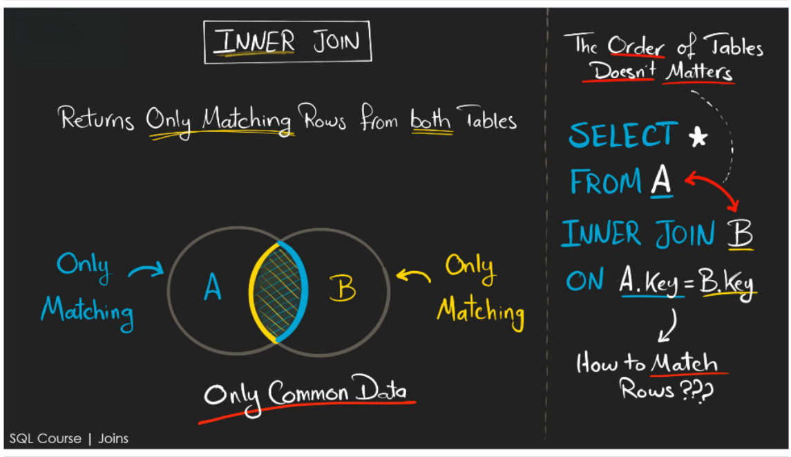

Returns Only Matching Rows from both Tables

```
SELECT *
FROM A
INNER JOIN B
ON A.Key = B.Key
```

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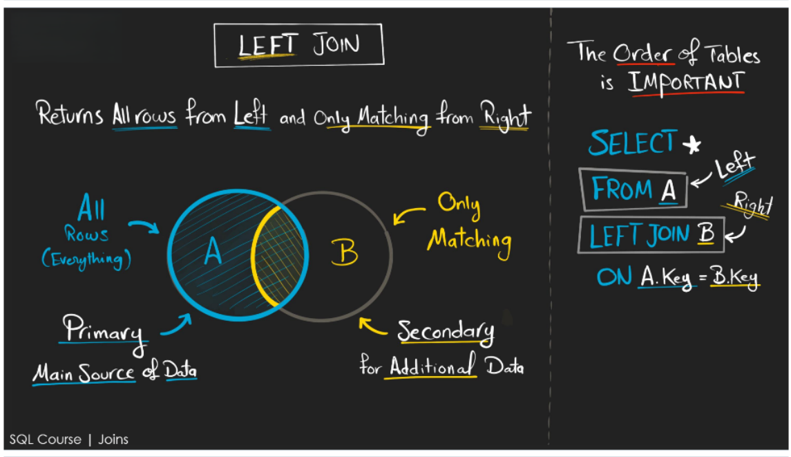

Returns All rows from Left and Only Matching from Right

```
SELECT *
FROM A
LEFT JOIN B
ON A.Key = B.Key
```

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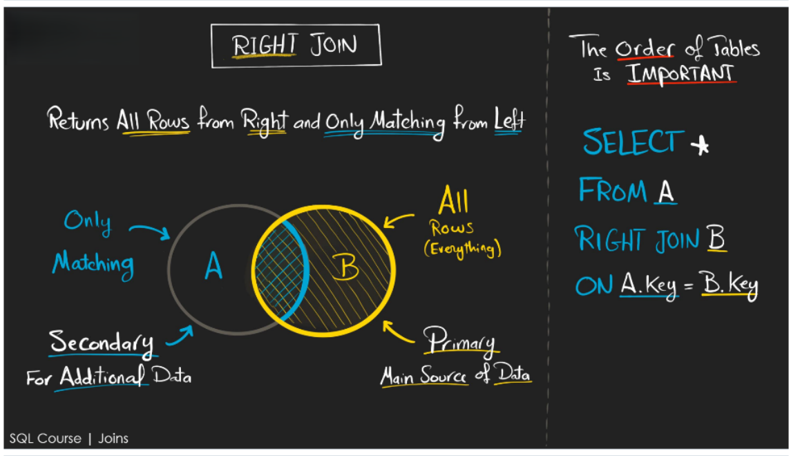

Returns All Rows from Right and Only Matching from Left

```
SELECT *
FROM A
RIGHT JOIN B
ON A.Key = B.Key
```

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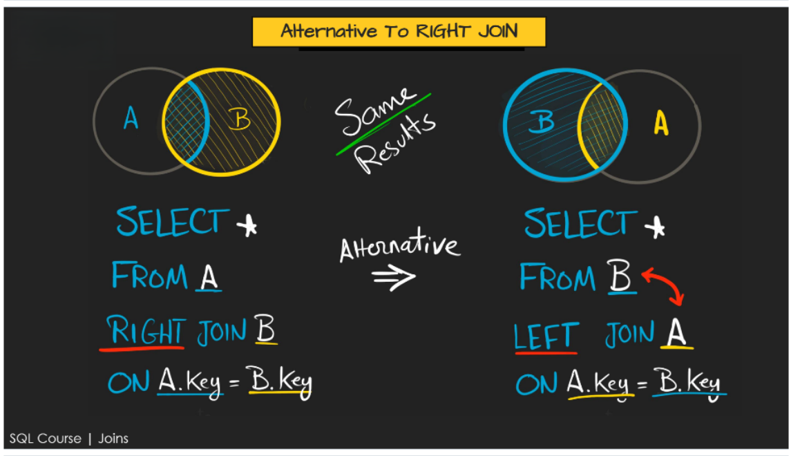

```
SELECT *
FROM A
RIGHT JOIN B
ON A.Key = B.Key
```

```
SELECT *
FROM B
LEFT JOIN A
ON A.Key = B.Key
```

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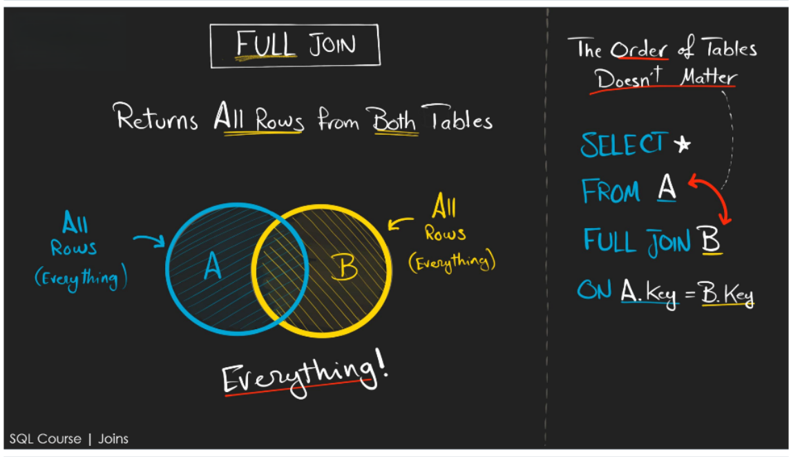

Returns All Rows from Both Tables

```
SELECT *
FROM A
FULL JOIN B
ON A.Key = B.Key
```

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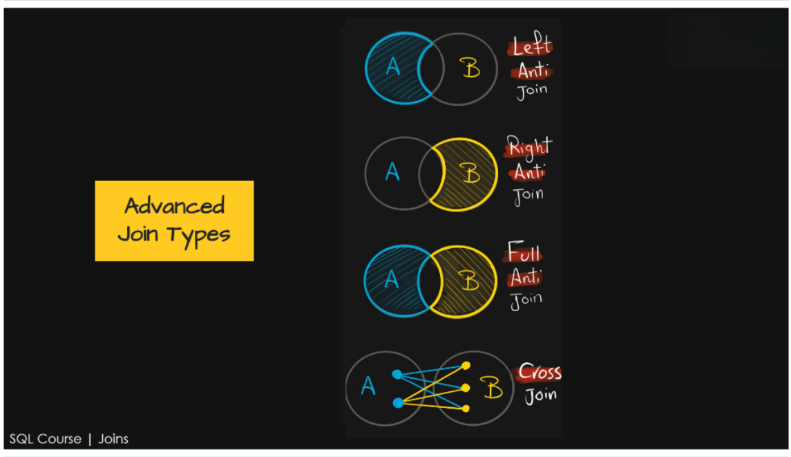

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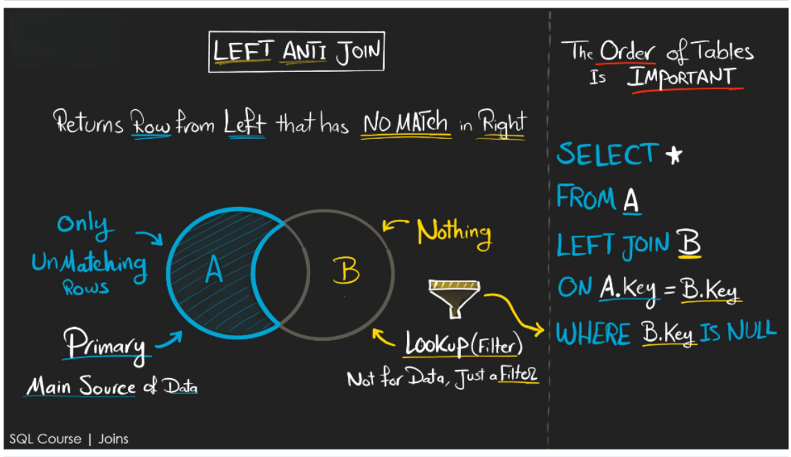

Returns Row from Left that has NO MATCH in Right

```
SELECT *
FROM A
LEFT JOIN B
ON A.Key = B.Key
WHERE B.Key IS NULL
```

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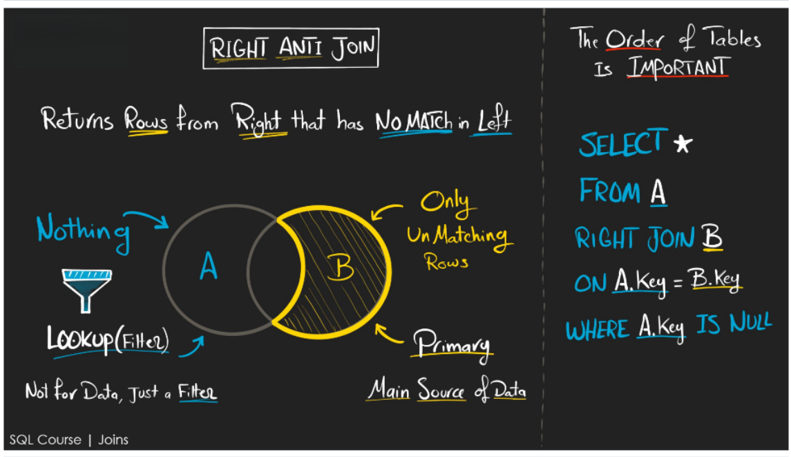

Returns Rows from Right that has NO MATCH in Left

```
SELECT *
FROM A
RIGHT JOIN B
ON A.Key = B.Key
WHERE A.Key IS NULL
```

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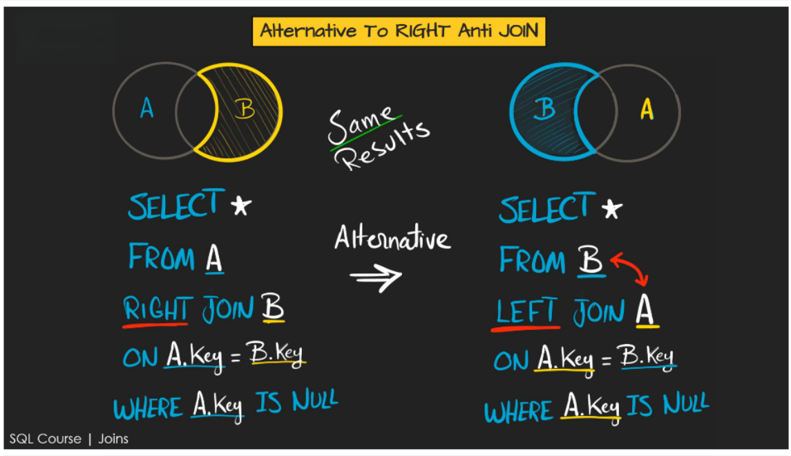

```
SELECT *
FROM A
RIGHT JOIN B
ON A.Key = B.Key
WHERE A.Key IS NULL
```

```
SELECT *
FROM B
LEFT JOIN A
ON A.Key = B.Key
WHERE A.Key IS NULL
```

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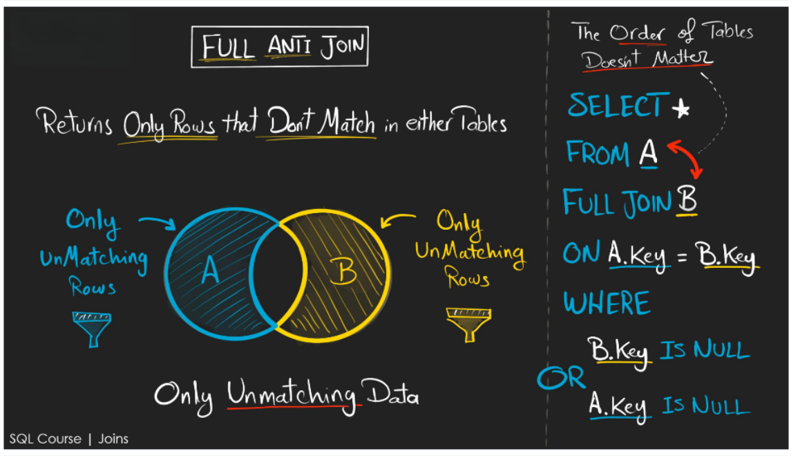

Returns Only Rows that Don't Match in either Tables

```
SELECT *
FROM A
FULL JOIN B
ON A.Key = B.Key
WHERE
   B.Key IS NULL
OR
   A.Key IS NULL
```

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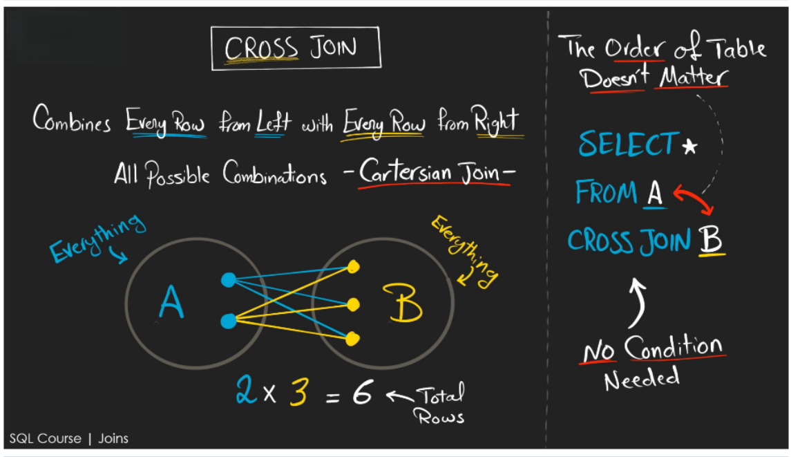

Combines Every Row from Left with Every Row from Right

All Possible Combinations - Cartersian Join -

```
SELECT *
FROM A
CROSS JOIN B
```

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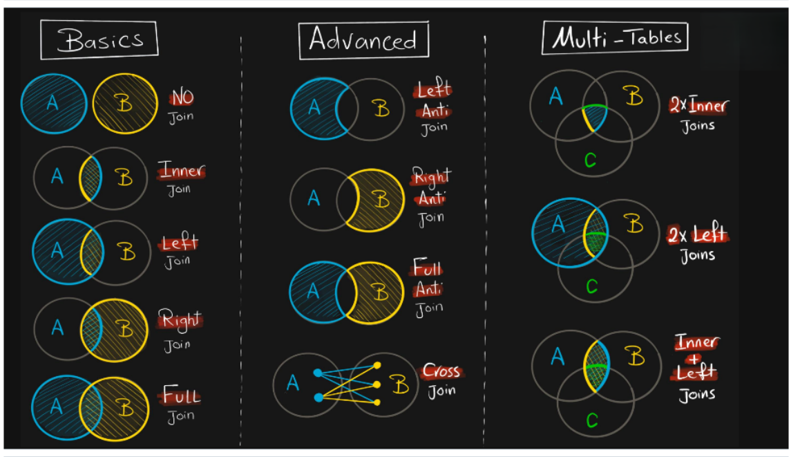

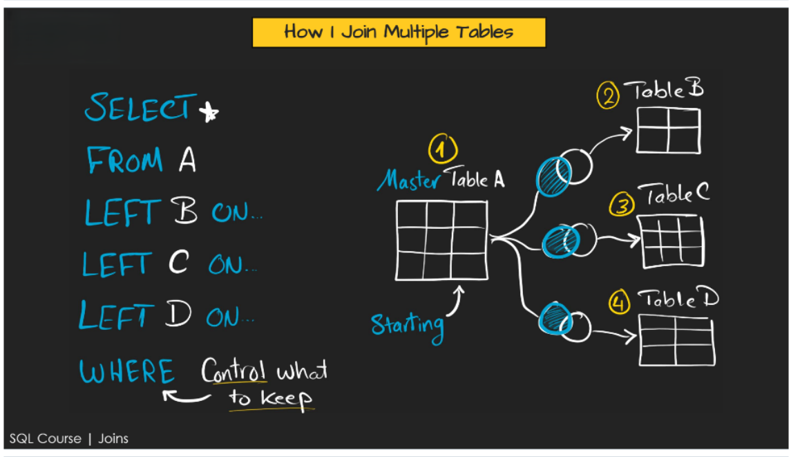

```
SELECT *
FROM A
LEFT B ON...
LEFT C ON...
LEFT D ON...
WHERE
```

Control what to keep

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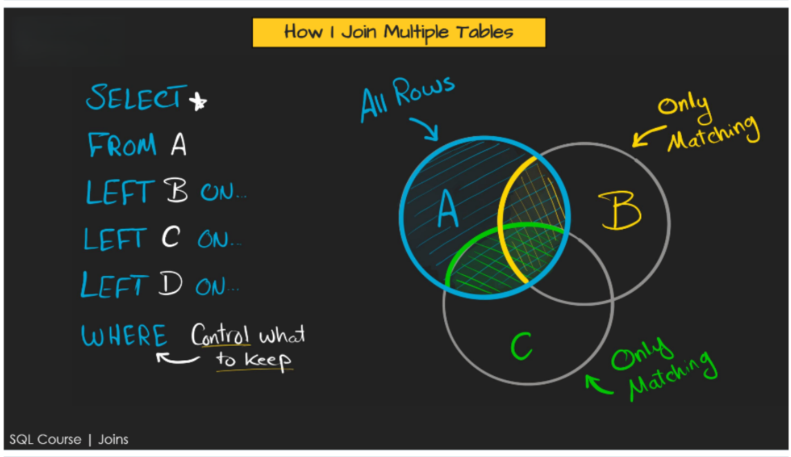

```
SELECT *
FROM A
LEFT B ON...
LEFT C ON...
LEFT D ON...
WHERE
```

Control what to keep

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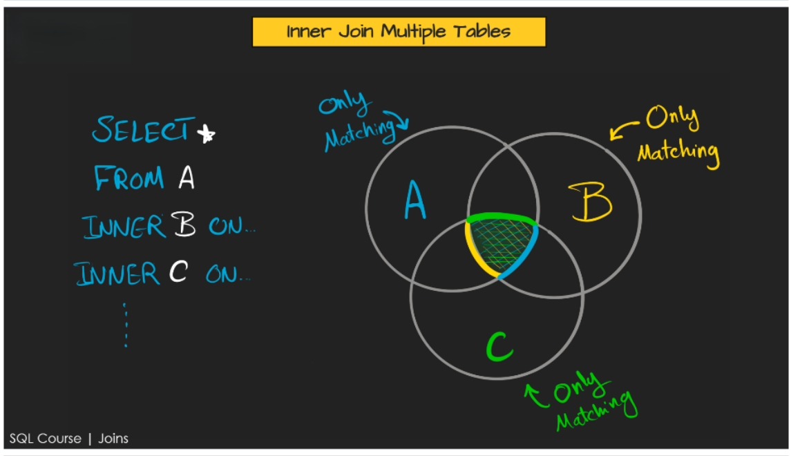


```
SELECT *
FROM A
INNER B ON...
INNER C ON...
⋮
```

SQL Course | Joins

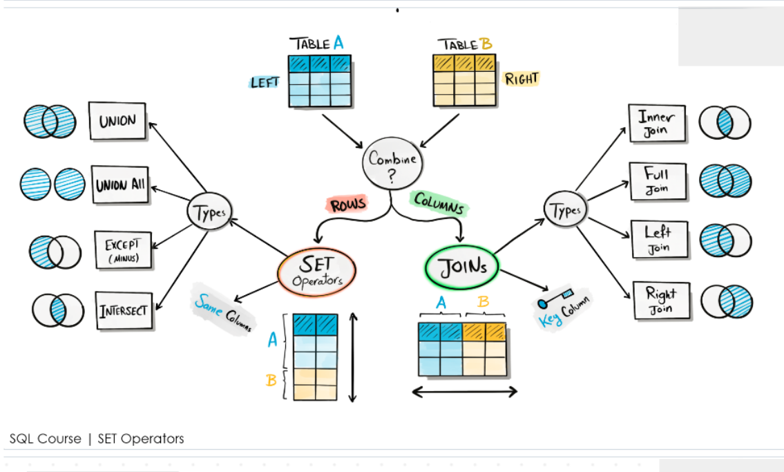
SET Operators

Combining Data

SET OPERATORS

Combine the results of multiple queries into a single result set.

SQL Course | SET Operators



SQL Course | SET Operators

CTE SYNTAX

```
SELECT
   FirstName
   LastName
FROM Customers
JOIN Clause
WHERE Clause
GROUP BY Clause
```

1st SELECT Statement

SET Operator

```
UNION
```

```
SELECT
   FirstName
   LastName
FROM Employees
JOIN Clause
WHERE Clause
GROUP BY Clause
```

2nd SELECT Statement

ORDER BY can be used only at the end to sort the final Result

```
ORDER BY FirstName
```

SQL Course | SET Operators

SET RULES

1. SET operators can be used in any clause.
2. ORDER BY is allowed only once—at the end of the query.
3. Each query must have the same number of columns.
4. Column data types must be compatible across queries.
5. The result set takes column names from the first query.

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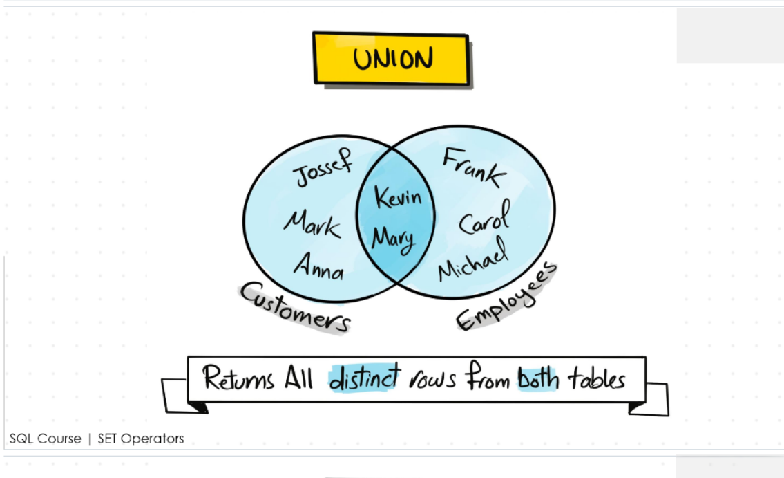

Returns All distinct rows from both tables

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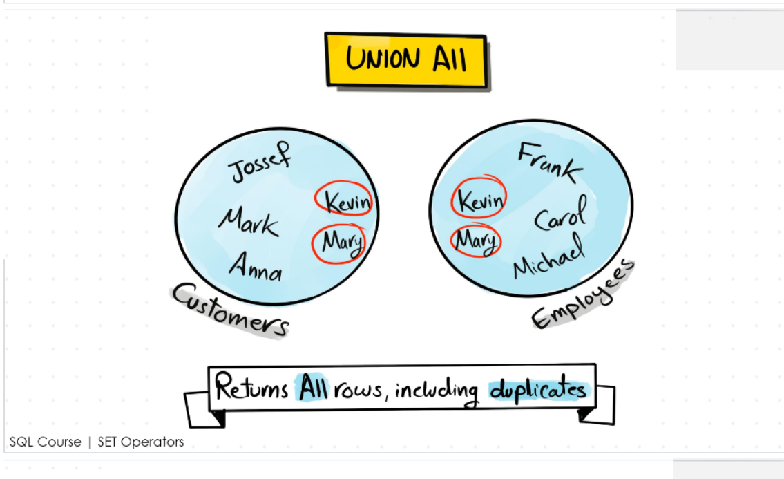

Returns All rows, including duplicates

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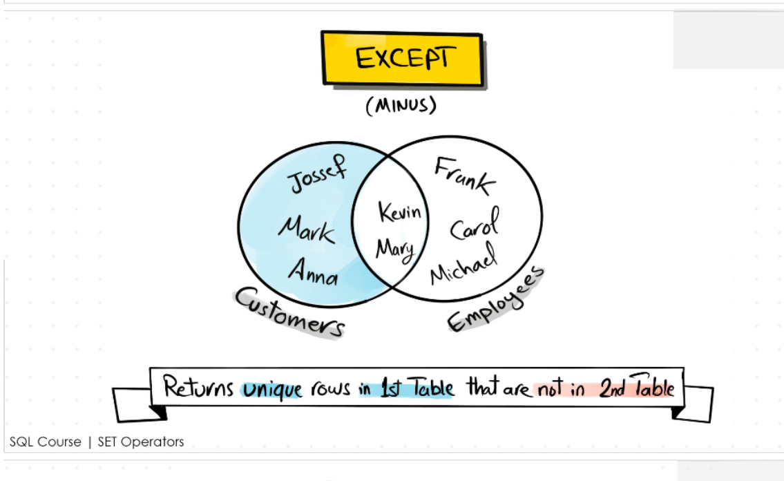

Returns unique rows in 1st Table that are not in 2nd Table

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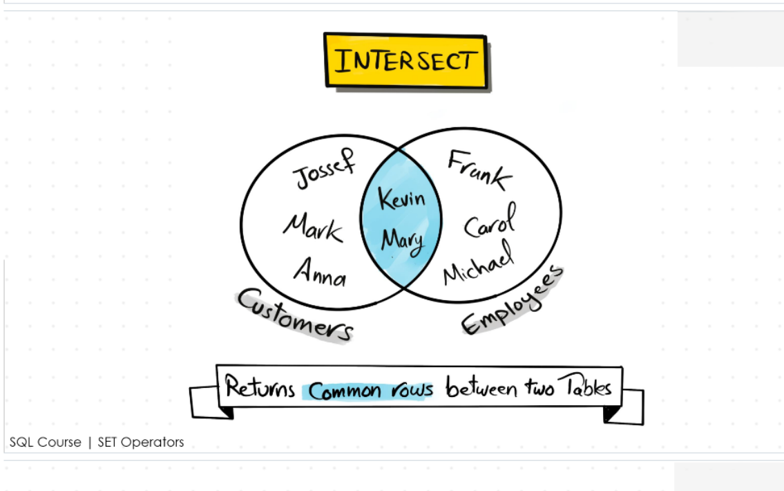

Returns Common rows between two Tables

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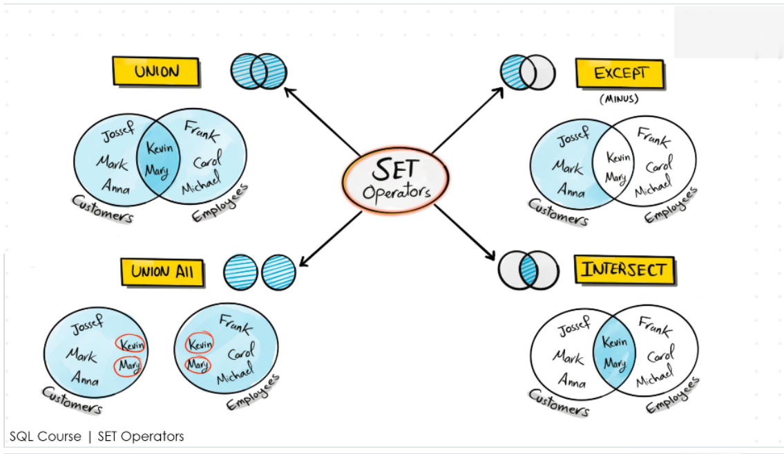

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UNION ALL vs UNION

- **UNION ALL** is faster than **UNION** because it doesn't remove duplicates.
- Use **UNION ALL** if you are sure there are no duplicates.
- Use **UNION ALL** to identify duplicates and data quality issues

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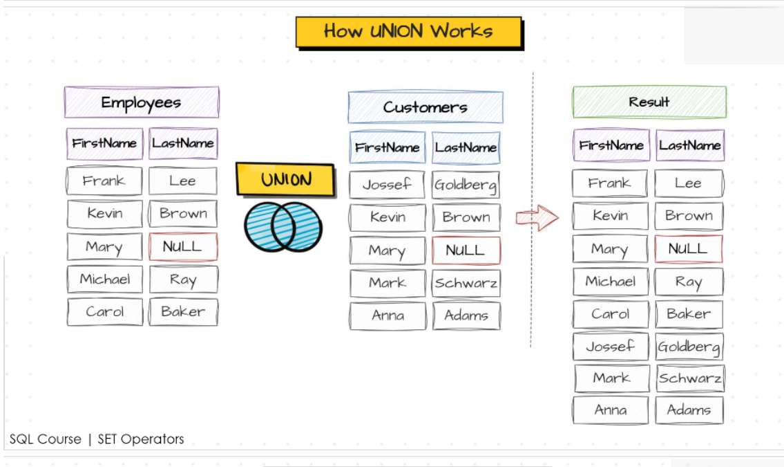


Employees

FirstName	LastName
Frank	Lee
Kevin	Brown
Mary	NULL
Michael	Ray
Carol	Baker

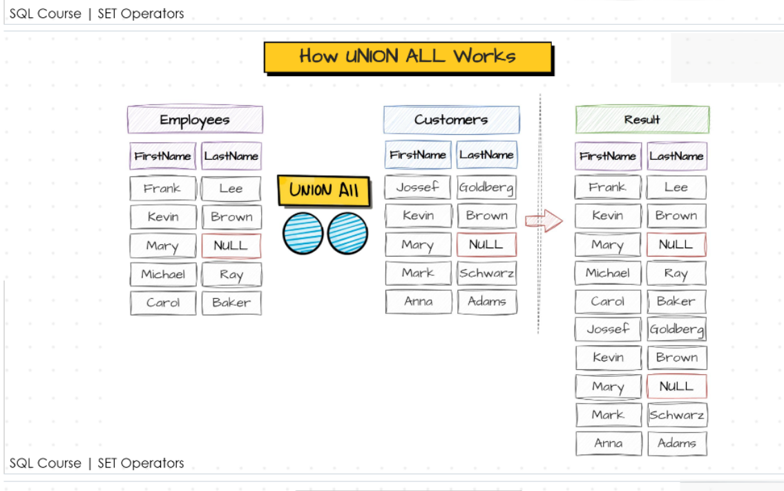
Customers

FirstName	LastName
Jossef	Goldberg
Kevin	Brown
Mary	NULL
Mark	Schwarz
Anna	Adams

Result

FirstName	LastName
Frank	Lee
Kevin	Brown
Mary	NULL
Michael	Ray
Carol	Baker
Jossef	Goldberg
Mark	Schwarz
Anna	Adams

SQL Course | SET Operators



Employees

FirstName	LastName
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Kevin	Brown
Mary	NULL
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Customers

FirstName	LastName
Jossef	Goldberg
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Mary	NULL
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Result

FirstName	LastName
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Mary	NULL
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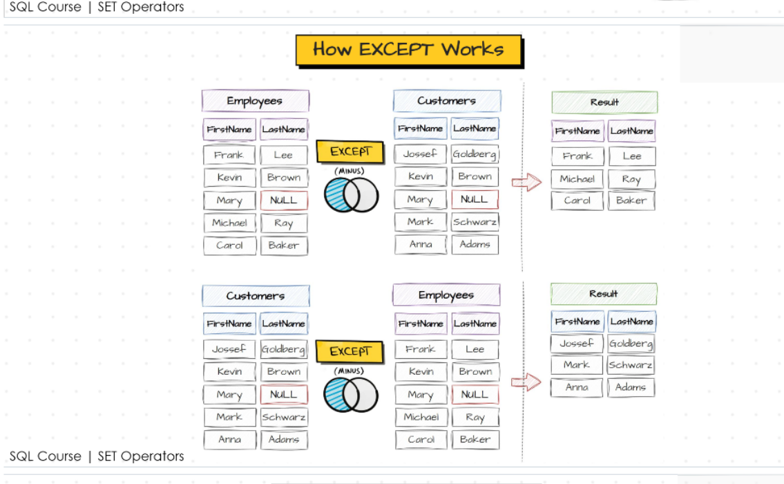
SQL Course | SET Operators



Employees

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Michael	Ray
Carol	Baker

Customers

FirstName	LastName
Jossef	Goldberg
Kevin	Brown
Mary	NULL
Mark	Schwarz
Anna	Adams

Result

FirstName	LastName
Frank	Lee
Michael	Ray
Carol	Baker

Customers

FirstName	LastName
Jossef	Goldberg
Kevin	Brown
Mary	NULL
Mark	Schwarz
Anna	Adams

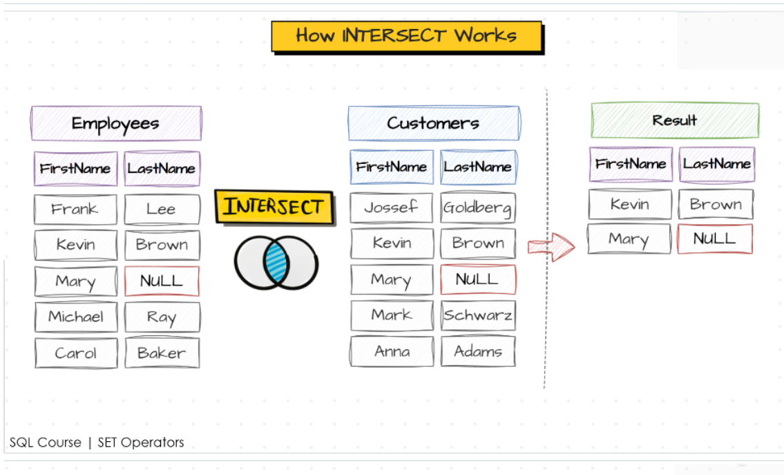
Employees

FirstName	LastName
Frank	Lee
Kevin	Brown
Mary	NULL
Michael	Ray
Carol	Baker

Result

FirstName	LastName
Jossef	Goldberg
Mark	Schwarz
Anna	Adams

SQL Course | SET Operators



Employees

FirstName	LastName
Frank	Lee
Kevin	Brown
Mary	NULL
Michael	Ray
Carol	Baker

Customers

FirstName	LastName
Jossef	Goldberg
Kevin	Brown
Mary	NULL
Mark	Schwarz
Anna	Adams

Result

FirstName	LastName
Kevin	Brown
Mary	NULL

SQL Course | SET Operators

SET OPERATORS

UNION	UNION ALL	EXCEPT/MINUS	INTERSECT
Returns All rows from both sets, elimination duplicates	Returns All rows from both sets, including duplicates	Return unique rows in first set that are not in second table	Return only the common rows between two sets

UNION

```
SELECT FirstName, LastName
FROM Customers

UNION

SELECT FirstName, LastName
FROM Employees
```

UNION ALL

```
SELECT FirstName, LastName
FROM Customers

UNION ALL

SELECT FirstName, LastName
FROM Employees
```

EXCEPT/MINUS

```
SELECT FirstName, LastName
FROM Customers

EXCEPT

SELECT FirstName, LastName
FROM Employees
```

INTERSECT

```
SELECT FirstName, LastName
FROM Customers

INTERSECT

SELECT FirstName, LastName
FROM Employees
```

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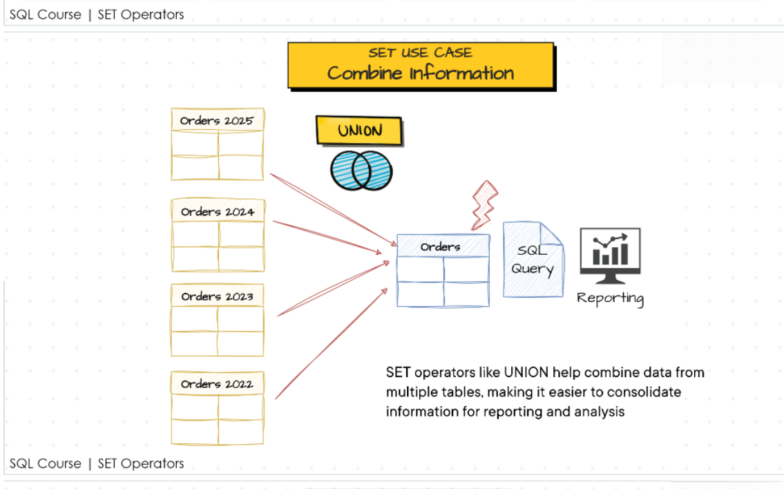


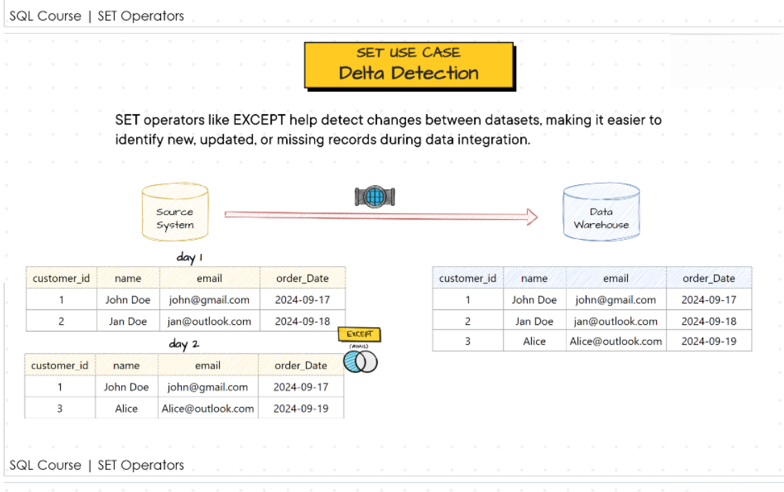
SET operators like UNION help combine data from multiple tables, making it easier to consolidate information for reporting and analysis

SQL Course | SET Operators

SET USE CASE

Delta Detection

SET operators like EXCEPT help detect changes between datasets, making it easier to identify new, updated, or missing records during data integration.



day 1

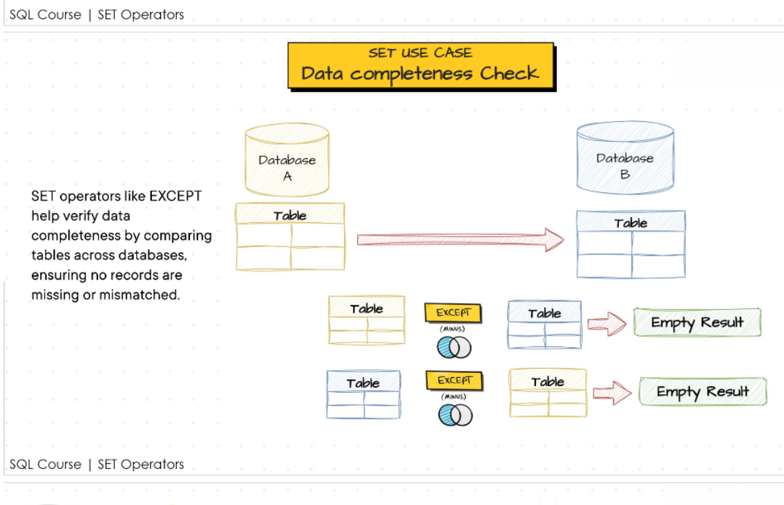
customer_id	name	email	order_Date
1	John Doe	john@gmail.com	2024-09-17
2	Jan Doe	jan@outlook.com	2024-09-18

day 2

customer_id	name	email	order_Date
1	John Doe	john@gmail.com	2024-09-17
3	Alice	Alice@outlook.com	2024-09-19

customer_id	name	email	order_Date
1	John Doe	john@gmail.com	2024-09-17
2	Jan Doe	jan@outlook.com	2024-09-18
3	Alice	Alice@outlook.com	2024-09-19

SQL Course | SET Operators



SET operators like EXCEPT help verify data completeness by comparing tables across databases, ensuring no records are missing or mismatched.

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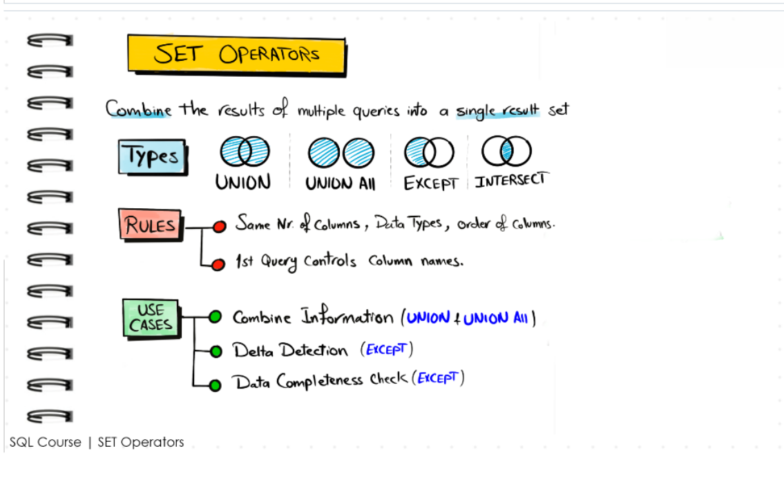

Combine the results of multiple queries into a single result set

Same Nr. of Columns, Data Types, order of columns.

1st Query Controls Column names.

Combine Information (UNION + UNION All)

Delta Detection (EXCEPT)

Data Completeness Check (EXCEPT)

SQL Course | SET Operators